

Sample Questions for Class Test II CITS2200 Data Structures and Algorithms

Q1. Which of the following statements is false for a doubly-linked list? Assume that you are given the reference **current** of an arbitrary node in the list, and each node stores two references **next** and **previous** to the node to its right and left

- (b) is false, as nodes in a doubly-linked list store an extra reference, that requires more space.*
- (a) Finding the **next** and **previous** references in a doubly-linked list is constant-time operations; (b) A doubly-linked list is more space-efficient compared to a singly-linked list; (c) The complexity of finding the reference of the last node in the list is $O(n)$; (d) Inserting a new node between the references **previous** and **current** takes $O(1)$ time.

Q2. Which of the following statements is false?

- this question is not included* *(c) is false, as time-complexity of heap sort is $O(n \log n)$*
- (a) The time complexity of selection sort is $O(n^2)$; (b) The time-complexity of insertion sort is $O(n^2)$; (c) The time-complexity of heap sort is $O(n^2)$; (d) Heap sort is an optimal sorting algorithm.

Q3. Which of the following statements is true for a min-heap?

- this question is not included* *(c) is true*
- (a) All the values in a subtree of a node k are smaller than the value at node k ; (b) The worst-case time complexity of restoring the heap property after inserting a new value is $O(n)$; (c) The values on a root to leaf path in the heap is in ascending order; (d) The worst-case time complexity of constructing a heap, given n integers is $O(n \log n)$.

Q4. Which of the followings is a valid representation of a single binary tree in an array. Assume that the root is stored in the location with array index 1.

(a)	45	15	24					34	21			5	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

(b)	45		15	7			24	34					15	5					
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

(c)	45	15	34					24	21			5	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

(d)	45	24	15	34	21					7	5								
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

(d) is correct

→ Q5. We now delete the number 7 from the binary tree in Q4. Which of the following is a valid representation of the binary tree now?

This question is not included

(a)	45	13	24					34	21			3								
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(b)	45			13				24	34					13	3					
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c)	45	24	13	34	21							3								
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(d)	45	13	34					24	21			3								
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c) is correct

Q6. Each node of a singly-linked list can store an item and a reference next to the node to the right. There are three nodes A, B and C in a linked-list such that A.next=B and B.next=C. Which of the following deletes the node B correctly?

(b) is correct.

- (a) A.next=B (b) A.next=A.next.next (c) A.next=B (d) B.next=A
B.Next=C

Q7. You are storing integers 26, 17, 14, 8, 5 in an array of size 5, using a hash function $(N + 6) \bmod 5$ when N is the integer you are storing. Which of the following is a correct representation of the array after storing these integers? The array index starts from 0.

you have to add 6 to each number and find the remainder by dividing by 5. The remainder is the array index.

(a)	17	26	14	8	5
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(b)	26	8	17	5	14
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(c)	14	5	26	17	8
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(d)	8	17	26	5	14
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(c) is correct

→ Q8. You are maintaining a priority queue using a linked list. You insert every element in the correct sorted position in the list. What are the worst-case times for enqueue and dequeue operations from this priority queue?

this question is not included

- (a) $O(1)$ and $O(n)$; (b) $O(n)$ and $O(\log n)$; (c) $O(n)$ and $O(1)$; (d) none of the above.

(c) is correct

→ Q9. You are given two min-heaps of n nodes each. You want to merge these two min-heaps by adding a root node and making the roots of the two

this question is not included

min-heaps as children of this new root node. What is the time complexity of merging these two min-heaps? *(c) is correct*

(a) $O(n \log n)$; (b) $O(n)$; (c) $O(\log n)$; (d) $O(n^2)$.

→ **Q10.** Given an unsorted array of n numbers, what is the complexity of the best algorithm using a heap for finding the minimum number in the array?

this question is not included

(a) $O(\log n)$; (b) $O(n \log n)$; (c) $O(n^2)$; (d) $O(n)$.

(d) is correct, as a min-heap can be constructed in $O(n)$ time